AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 25 · STUDENT LESSON

Metrics Match Consequences

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Week 17: precision and recall use different denominators.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Apply a declared score threshold including equality. Try the recall before opening notes.

Your learning target

I can apply a declared **score threshold** including equality.

I can compare **false alarm**s and misses at two **threshold**s.

I can separate validation choices from final testing.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a **score** for your ability.

Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Metric choice depends on intended use, the positive class, who is affected, and the **consequence**s of each error.



Your target

LEARNING TARGET A reviewer can inspect at most 10 alerts daily. Model A flags 20 cases with higher recall; B flags 8 with lower recall. What **evidence** is needed before choosing a **threshold**?

Builder name and experiment date:

Readiness for the connected explanation

Which denominator defines recall?

This week's end goal

By the end, I can explain metrics match **consequence**s using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

Investigate the supplied Metrics Match Consequences case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use metrics match **consequence**s to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A classifier **score** is not automatically a probability. Declare blocked positive and predict blocked when **score** is at least the **threshold**. Changing this boundary changes decisions without necessarily re**training** the model.

Select it using development **evidence** and a task-relevant rule, not final-test answers.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **threshold**s and **consequence**s.

STUDENT LESSON · 3 OF 18

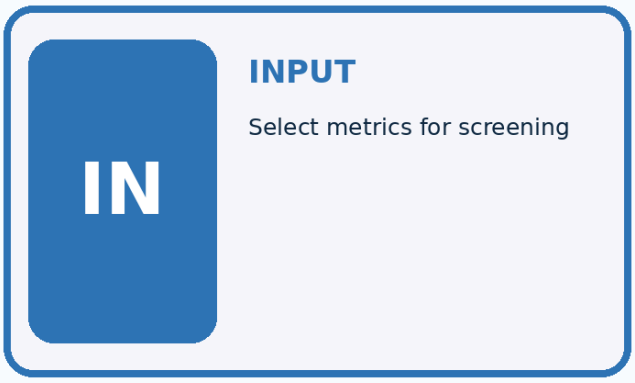
See how it works

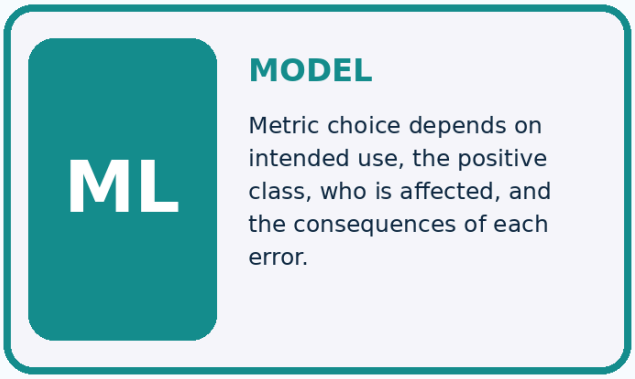
> Follow the mechanism for Metrics Match Consequences. Connect each labelled input, deciding step and result.

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Select metrics for screening, recommendation, and forecasting tasks.

MODEL IDEA Metric choice depends on intended use, the positive class, who is affected, and the **consequence**s of each error.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Screening task treats missed positives as costly, so emphasize recall and human follow-up. Recommendation task may track precision among shown items plus user benefit.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Six records have **score**/reference pairs A .9/blocked, B .7/clear, C .6/blocked, D .4/blocked, E .3/clear, F .1/clear. At **threshold** .5, TP=2, FP=1, FN=1, TN=2. At .3, TP=3, FP=2, FN=0, TN=1.

Equality is positive, so E is flagged at .3.

```
scores = [0.9, 0.7, 0.6, 0.4, 0.3, 0.1]
actual = [True, False, True, True, False, False]
for threshold in [0.5, 0.3]:
    predicted = [s >= threshold for s in scores]
    fp = sum(p and not a for p, a in zip(predicted, actual))
    fn = sum(not p and a for p, a in zip(predicted, actual))
    print(threshold, fp, fn, fp + 4 * fn)
# Output rows: 0.5 1 1 5, then 0.3 2 0 2.
```

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

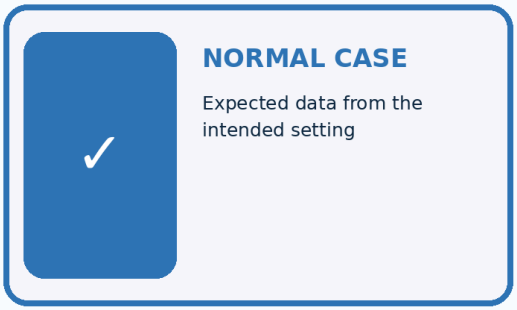
Compare the cases

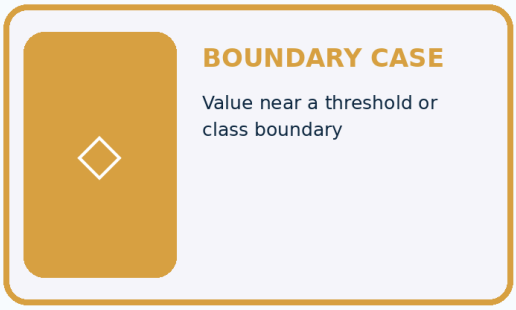
> Compare the supplied cases for Metrics Match Consequences. Identify the detail that changes the conclusion.

EVALUATION DESIGN

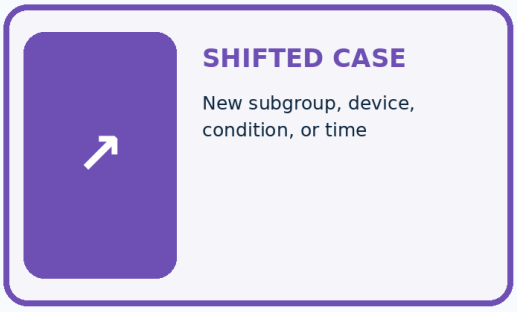
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a **threshold**, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: Miss and false-alarm costs swap to1 and 5.

Predicted result

Prediction: **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches.

Why the result follows

Reason: Metric choice encodes the declared **consequence**s, not a universal best **threshold**.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Metrics Match Consequences to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

CONSEQUENCE

real effect of a prediction error

2

THRESHOLD

score boundary for a class decision

3

COST MATRIX

relative impact assigned to error types

4

METRIC SUITE

multiple measures covering distinct concerns

Term	Precise meaning	My ML example
consequence	real effect of a prediction error	
threshold	score boundary for a class decision	
cost matrix	relative impact assigned to error types	
metric suite	multiple measures covering distinct concerns	

MISCONCEPTION TO REPAIR The same highest-accuracy metric should choose every model.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Score	A model output used to rank or decide
Threshold	A boundary that converts a score to a label
False alarm	A positive prediction for a negative reference
Cost rule	A declared way to value different mistakes

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Metrics Match Consequences, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 25 • WORKED EVIDENCE

1. Screening task treats missed positives as costly, so emphasize recall and human follow-up.
2. Recommendation task may track precision among shown items plus user benefit.
3. Height forecast uses MAE in centimeters and residual slices.
4. No metric replaces a written consequence analysis and subgroup review.

Step 1

STEP 1 Screening task treats missed positives as costly, so emphasize recall and human follow-up.

Step 2

STEP 2 Recommendation task may track precision among shown items plus user benefit.

Step 3

STEP 3 Height forecast uses MAE in centimeters and residual slices.

Step 4

STEP 4 No metric replaces a written **consequence** analysis and subgroup review.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a **score**.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Thresholds and consequences at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Threshold .5	One false alarm one miss	Four of six correct
Threshold .3	Two false alarms no misses	Four of six correct
Decision rule	Include equality	Record before evaluation

Lowering the threshold here finds D but adds E as a false alarm. The two accuracies are equal while consequences differ. Precision is 2/3 at .5 and 3/5 at .3; recall rises from 2/3 to 1. This behavior follows these fixed scores and records, not a promise of future performance.

Give TP, FP, FN, TN at both thresholds. Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Metrics Match Consequences

Translate a **threshold** choice into missed-leaf and false-alarm costs.

The facts we will use

Supplied facts

Scores/reference: 0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy. Predict spotted at **score**>=**threshold**. Miss cost5; false-alarm cost1.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/25

Metrics Match Consequences

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Threshold0.7

TP 1, FN 1, FP 0, TN 2: cost5.



2. Threshold0.4

TP 2, FN 0, FP 1, TN 1: cost1.



3. Choose for stated costs

Threshold0.4 has smaller validation cost on these four cases.

CHANGE ONE THING

Miss and false-alarm costs swap to1 and 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Threshold0.7	TP 1, FN 1, FP 0, TN 2: cost5.
2. Threshold0.4	TP 2, FN 0, FP 1, TN 1: cost1.
3. Choose for stated costs	Threshold0.4 has smaller validation cost on these four cases.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. **Threshold**0.4 has smaller validation cost on these four cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Toy costs illustrate decision design; real use needs agreed **consequence**s, review capacity and fresh evaluation.

Keep the stated limit

Read the labelled picture: Which labels determine the false-alarm count at **threshold** 0.4? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Miss and false-alarm costs swap to1 and 5. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Metrics Match Consequences claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: The same highest-accuracy metric should choose every model. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE The same highest-accuracy metric should choose every model.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

These invented costs support arithmetic practice, not real robot deployment. Stakeholders must justify actual **consequence**s.

Record every tried **threshold** and retain a separate final test; a **threshold** is part of the selected method.

Why not select the threshold with the lowest final-test cost?

STUDENT LESSON · 8 OF 18

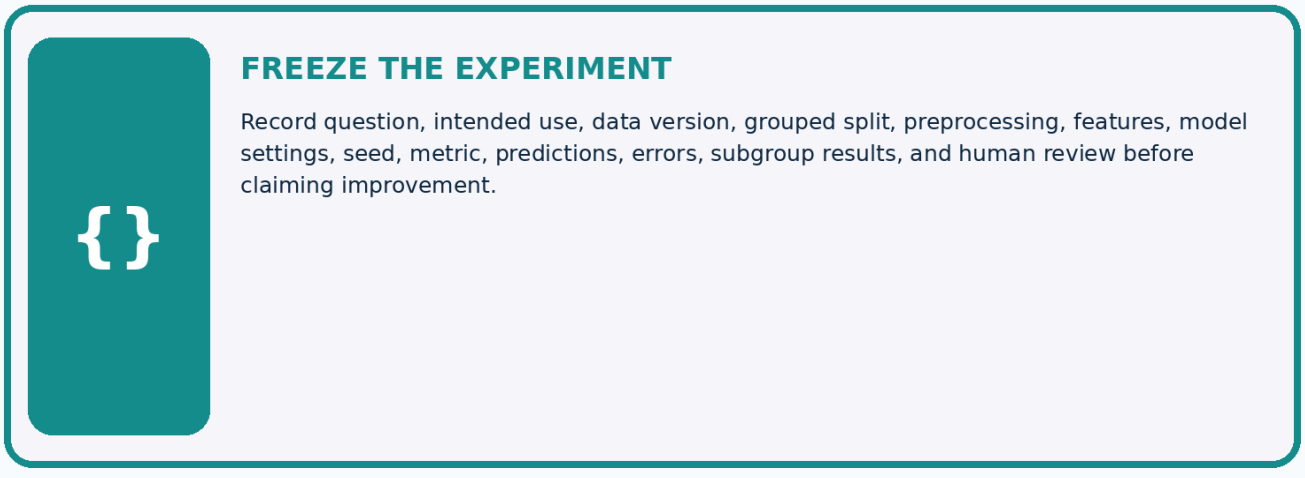
Practise together

Practise Metrics Match Consequences with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

> **Question to explain**
>
> QUESTION + intended use + success **threshold**

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Suppose the classroom cost is one point per **false alarm** and four per miss. Total costs are 5 at .5 and 2 at .3, so .3 wins among these two candidates under this rule.

If **false alarm**s cost four and misses cost one, costs are 5 and 8, reversing the choice.

Practise with less help: Count the consequences

Use workbook W2 for the connected example "**Threshold**s and **consequence**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Threshold**s and **consequence**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which labels determine the false-alarm count at **threshold** 0.4? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Metrics Match Consequences. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Metric Council: map stakeholders and error costs, select a metric suite, and defend what each metric cannot show.

1. Write the hypothesis and expected **evidence** before running.

2. Metric Council: map stakeholders and error costs, select a **metric suite**, and defend what each metric cannot show.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

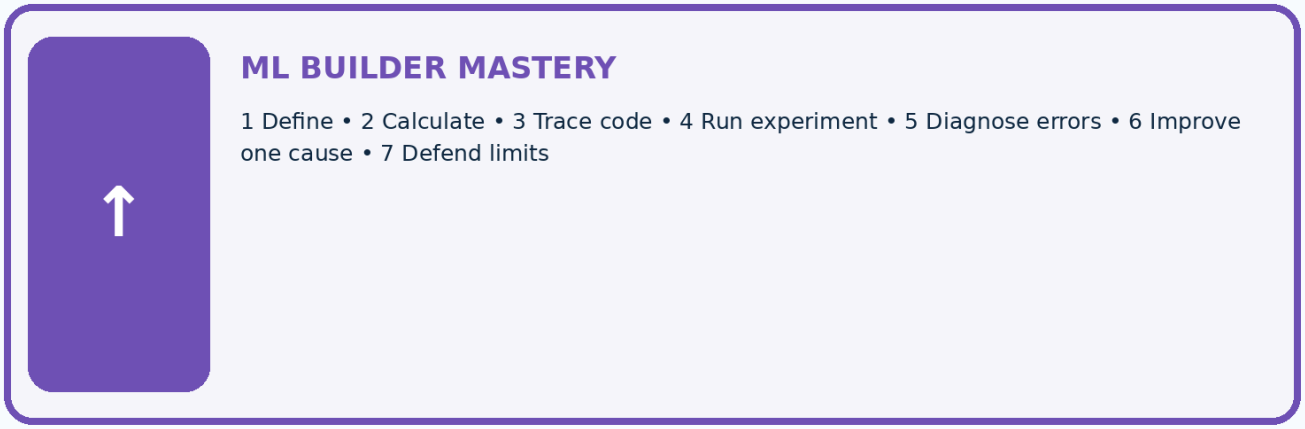
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Metrics Match Consequences. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **consequence**, **threshold**, **cost matrix**, **metric suite**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE The same highest-accuracy metric should choose every model.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

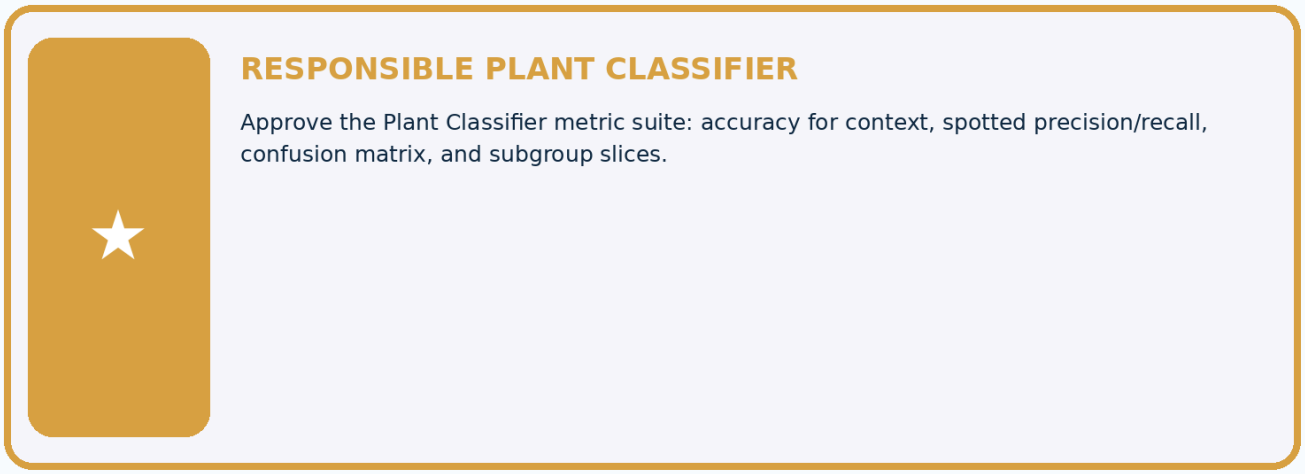
Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Approve the Plant Classifier **metric suite**: accuracy for context, spotted precision/recall, confusion matrix, and subgroup slices.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

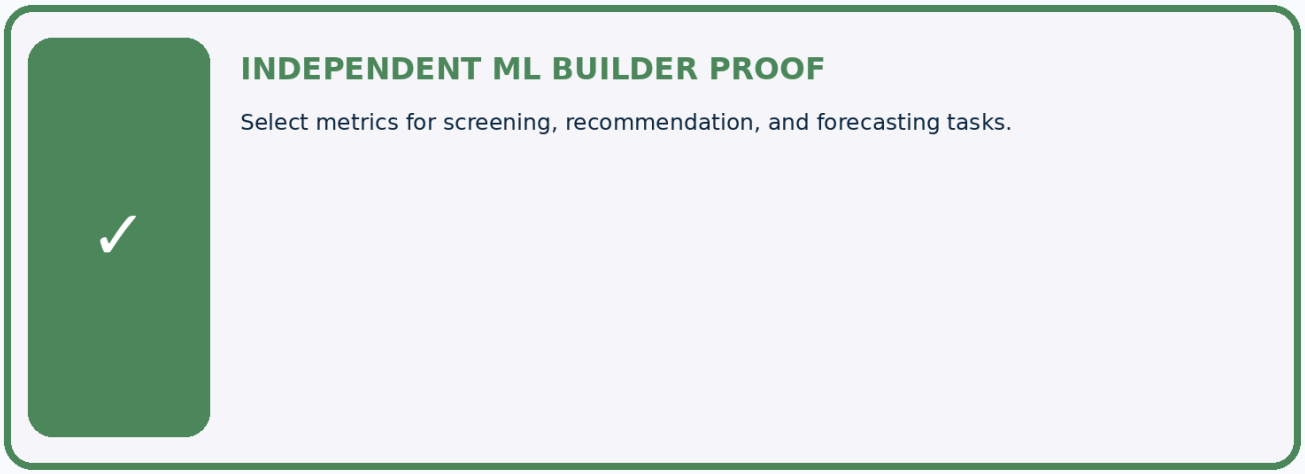
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Select metrics for screening, recommendation, and forecasting tasks.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Reproducible Experiments.

Study a complete case

> Revisit the complete Metrics Match Consequences case for supported practice. Repeated examples do not become fresh assessments.

Week 25 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Spotting alerts can miss disease signs or create unnecessary healthy-leaf reviews.

The question

Connect metrics to **consequence**s.

Worked explanation

Recall measures finding positives; precision measures alert correctness. Set tradeoffs using **consequence**s, review capacity and validation **evidence**.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Connect error types to consequences

Evidence or stage	Value or result
False negative	Spotted leaf missed
False positive	Healthy leaf creates an alert
Recall	Fraction of positives found
Precision	Fraction of alerts correct

Spotting alerts can miss disease signs or create unnecessary healthy-leaf reviews.

Connect metrics to **consequence**s.

Recall measures finding positives; precision measures alert correctness. Set tradeoffs using **consequence**s, review capacity and validation **evidence**.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

Scores/references are .8/blocked,.5/clear,.2/blocked,.1/clear. Compare **threshold**s .5 and .2 using the same cost.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 25 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> Missing a diseased plant is considered more costly than a **false alarm**, but staff can review only ten alerts daily. Which evaluation trade-off should be discussed?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Approve the Plant Classifier **metric suite**: accuracy for context, spotted precision/recall, confusion matrix, and subgroup slices.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A reviewer can inspect at most 10 alerts daily. Model A flags 20 cases with higher recall; B flags 8 with lower recall. What **evidence** is needed before choosing a **threshold**?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Metrics Match Consequences, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Translate a **threshold** choice into missed-leaf and false-alarm costs.

By the end, I can	How I will show it
Apply a declared score threshold including equality	Explain the deciding step, show a trace or test, and state one limit.
Compare false alarms and misses at two thresholds	Explain the deciding step, show a trace or test, and state one limit.
Separate validation choices from final testing	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Metrics Match Consequences without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Apply a declared **score threshold** including equality

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Translate a **threshold** choice into missed-leaf and false-alarm costs.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? **Threshold**0.4 has smaller validation cost on these four cases.

1. **Threshold**0.4

2. Choose for stated costs

3. **Threshold**0.7

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Choose for stated costs **Threshold**0.4 has smaller validation cost on these four cases. Read the picture in order.

Threshold0.4 has smaller validation cost on these four cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Miss and false-alarm costs swap to1 and 5. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches. Metric choice encodes the declared **consequence**s, not a universal best **threshold**.

Repair a missing building block

Review Train, Validation, Test (ai-4/24). Goal: Select a model using a declared validation rule. Short practice: Why does B's zero **training** MAE not settle the original choice?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train a Regression Model (ai-4/22). Goal: Fit a line from a declared **training** candidate set. Short practice: May day3's evaluation label choose the candidate line?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Regression and Residuals (ai-4/19). Goal: Distinguish numeric **prediction** from classification. Short practice: Which statistic reveals the size of errors without cancellation?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

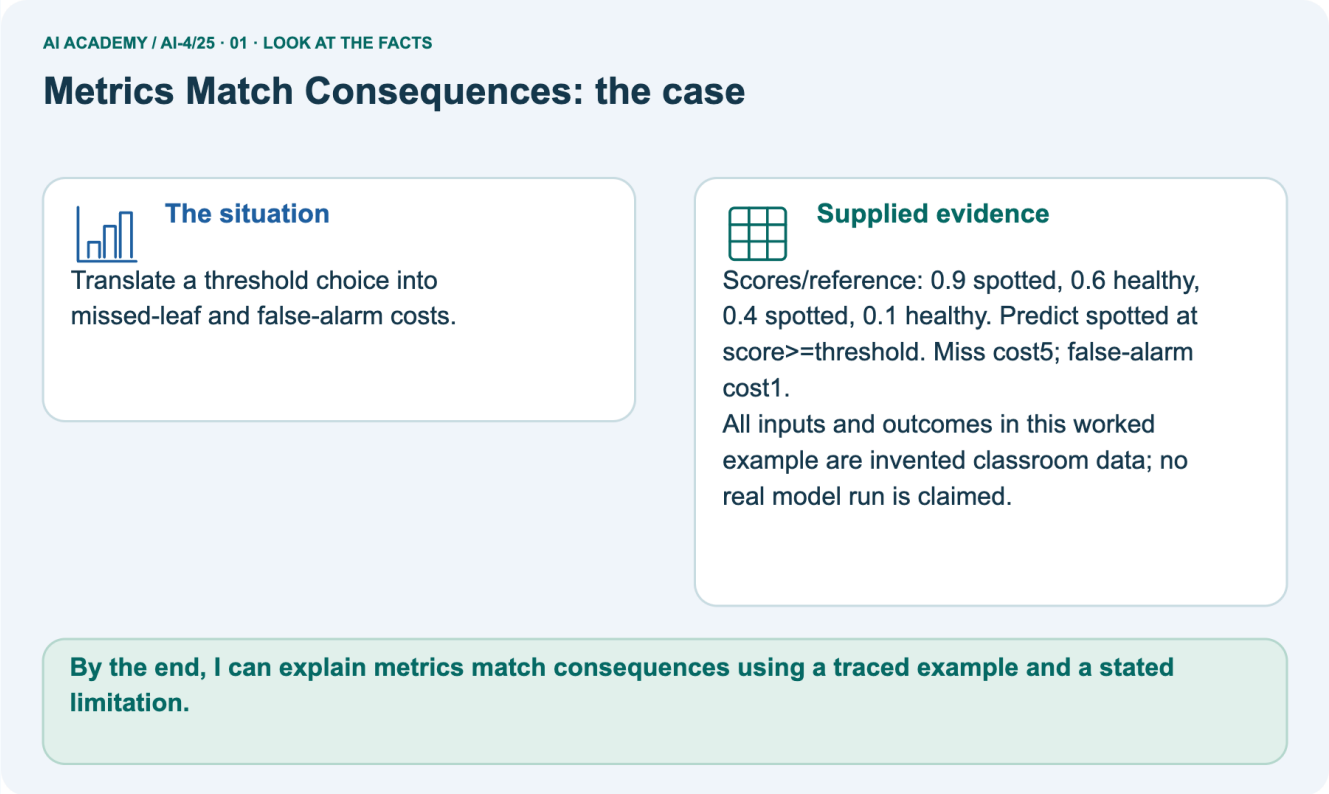
Drafts stay in this browser when storage is available; clear drafts on a shared device.

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Visual concept journey

> Explain each Metrics Match Consequences graphic using its labels, deciding evidence and relevant limitation.

01. Metrics Match Consequences: the case



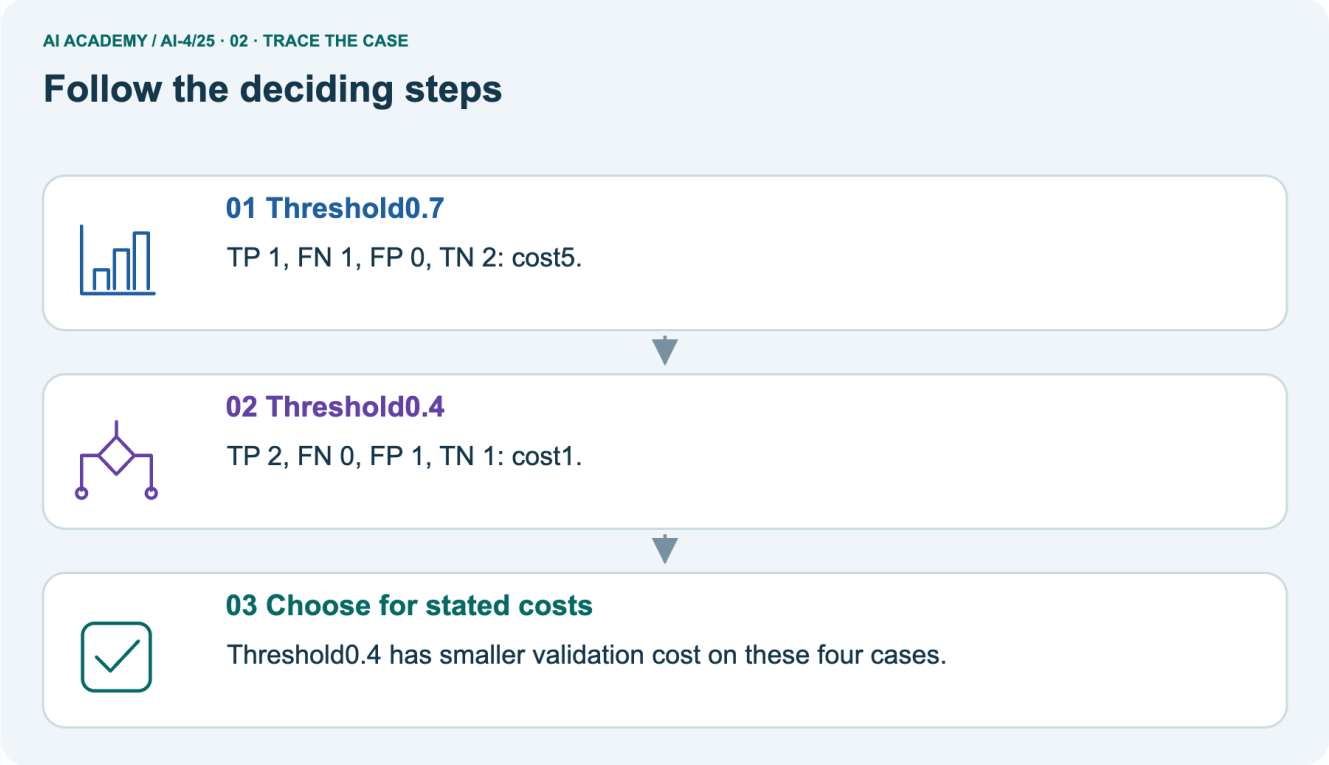
Follow the labelled picture with the matching explanation. Open the vector diagram

Translate a **threshold** choice into missed-leaf and false-alarm costs. Supplied facts: **Score**s/reference: 0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy. Predict spotted at **score**>=**threshold**. Miss cost5; false-alarm cost1.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

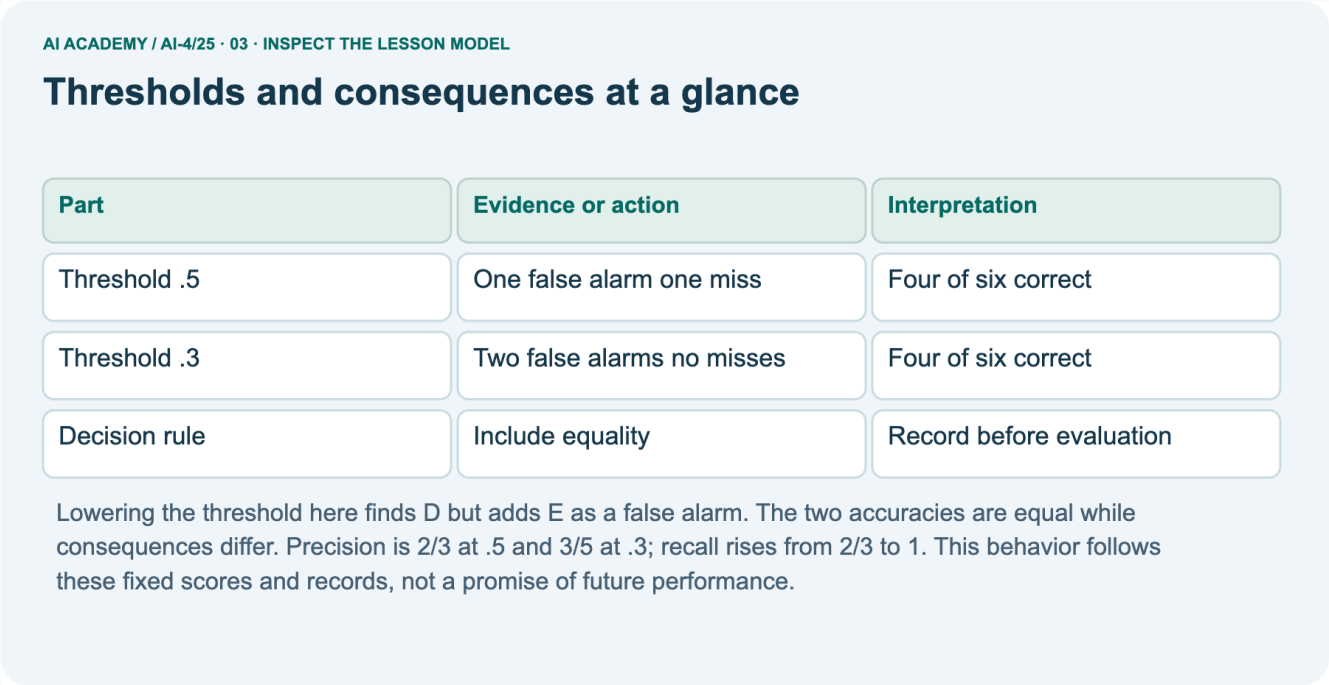


Follow the labelled picture with the matching explanation. Open the vector diagram

Threshold0.7: TP 1, FN 1, FP 0, TN 2: cost5. → **Threshold**0.4: TP 2, FN 0, FP 1, TN 1: cost1. → Choose for stated costs: **Threshold**0.4 has smaller validation cost on these four cases.

Point to each numbered stage and explain which recorded **input** it uses.

03. Thresholds and consequences at a glance



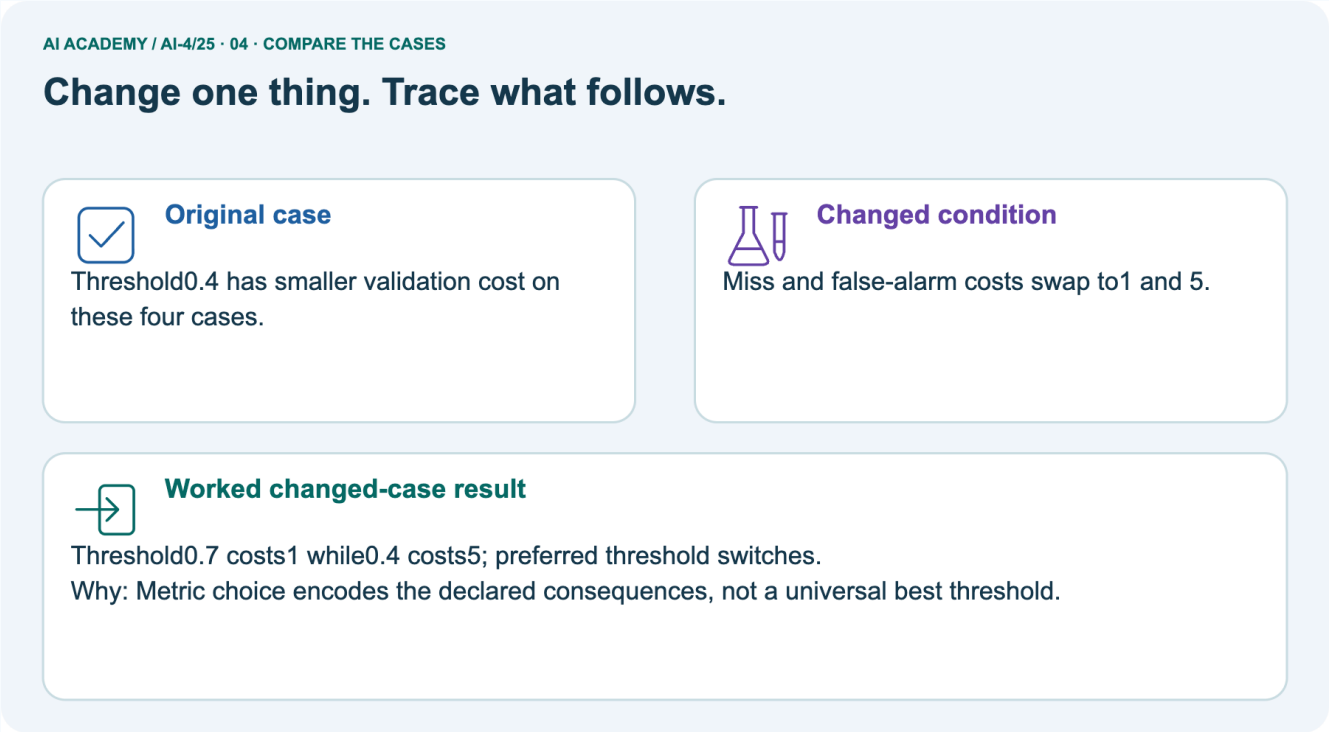
Follow the labelled picture with the matching explanation. Open the vector diagram

Thresholds and **consequence**s at a glance. **Threshold** .5 | One **false alarm** one miss | Four of six correct. **Threshold** .3 | Two **false alarm**s no misses | Four of six correct.

Decision rule | Include equality | Record before evaluation

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



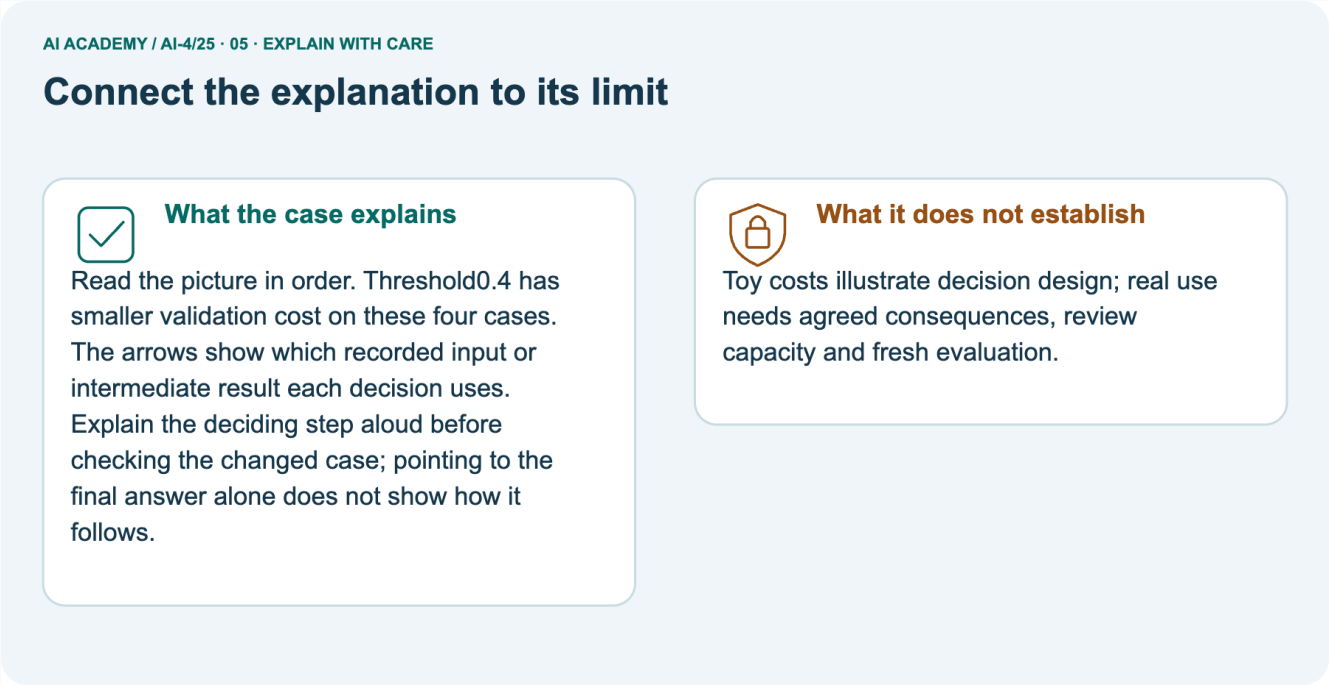
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Threshold**0.4 has smaller validation cost on these four cases. Changed condition: Miss and false-alarm costs swap to1 and 5. Expected result: **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches.

Reason: Metric choice encodes the declared **consequence**s, not a universal best **threshold**.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

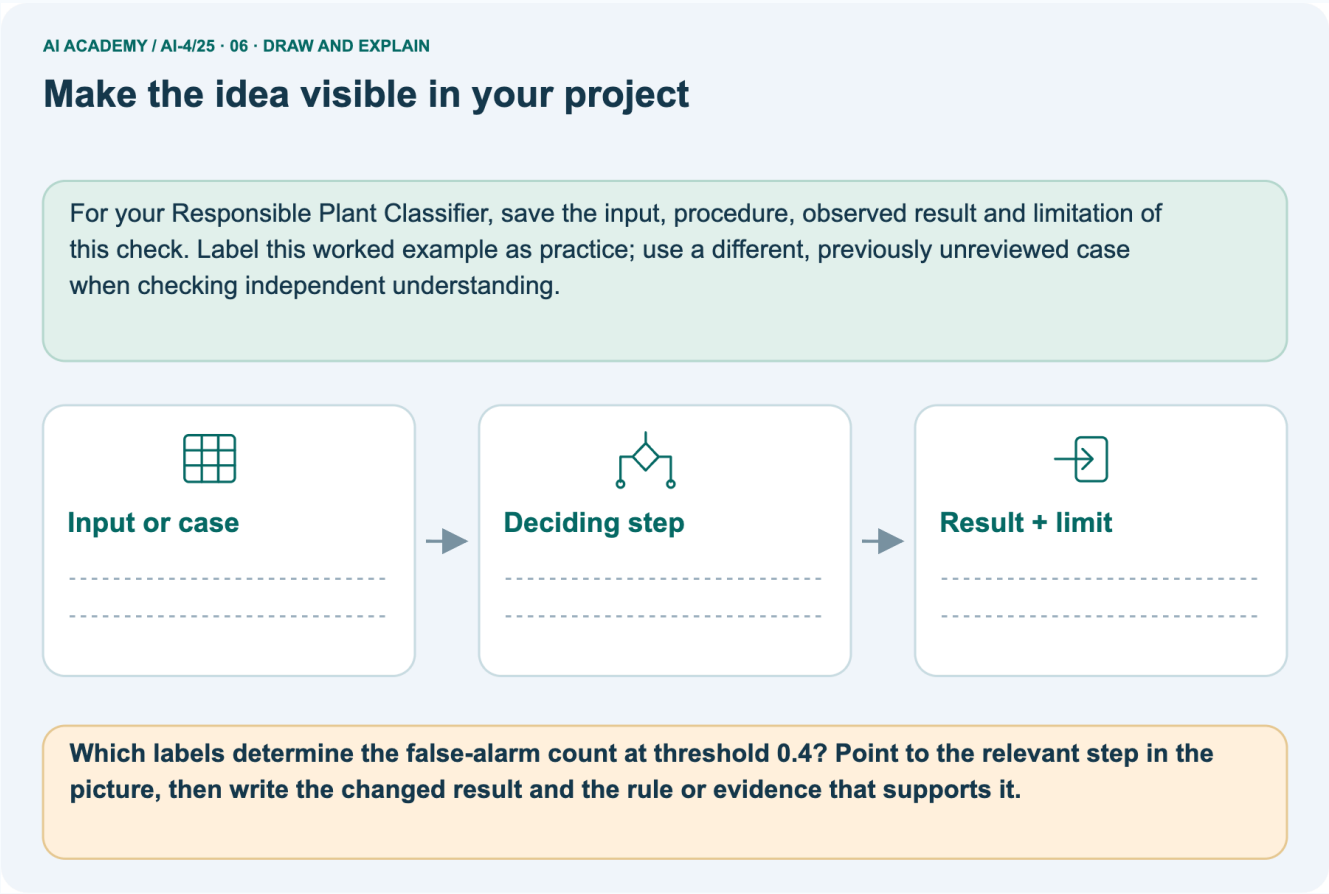
Read the picture in order. **Threshold**0.4 has smaller validation cost on these four cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Toy costs illustrate decision design; real use needs agreed **consequence**s, review capacity and fresh evaluation.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which labels determine the false-alarm count at **threshold** 0.4? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.